Department of Electrical and Information Engineering

University of Ruhuna

Mini Project Proposal

Image Processing and Computer Vision

EE7204 / EC7205

Project Title

Student Name 1

Student Name 2

Student Name 3

Student Name 4

Submission date

Department of Electrical and Information Engineering
University of Ruhuna

Contents

1. Problem Statement

(Maximum word limit: 150 words) Clearly define the problem or motivation behind your project. Explain and justify what issue, limitation, or real-world challenge you aim to address through image processing or computer vision with existing work.

How to use references using BibTeX files [1].

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2. Objectives

List the main goals of your mini project. Highlight what you intend to achieve or demonstrate at the end of the project. Give points wise.

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3. Literature Review

(Minimum word limit: 1000 words) Briefly summarize 3–6 previous research works, projects, or academic papers related to your topic. Discuss what methods they used, their strengths and limitations, and how your proposed project will differ from or improve upon them.

Example structure:

- **Study 1:** Summary of method and result.
- **Study 2:** Summary of technique and limitation.
- **Gap:** Highlight what your project adds beyond existing work.

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4. Methodology

(Minimum word limit: 1000 words) Describe the approach, tools, and algorithms you will use. Mention the main image processing or computer vision techniques (e.g., segmentation, feature extraction, classification) and the software frameworks (e.g., OpenCV, MATLAB, TensorFlow).

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5. Dataset / Data Collection

(Maximum word limit: 100 words) Provide details of the data its source, number of images, format, and characteristics. If you plan to collect your own data, explain how you will ensure data quality, diversity, and ethical use.

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6. Expected Outcomes

(Maximum word limit: 100 words) Summarize the expected outputs (eq: demo, code files, presentation) and performance metrics (e.g., accuracy, clarity, speed). Explain how your results will demonstrate the effectiveness of your image processing or computer vision approach. Write the outcomes in point form.

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7. Percentage of Image Processing & Computer Vision Involvement

(Approximately 50–100 words) State clearly how your project satisfies the requirement of at least 65% image processing and computer vision components. For example, describe which parts involve preprocessing, segmentation, or model development etc. And if you are planning to use any tools may be you can mention.

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References

[1] R. C. Gonzalez and R. E. Woods, "Digital image processing," *Prentice Hall*, 2008.

A minimum of ten (10) references must be included in the References section.